

Longitudinal Gaps and Spatial Hit Connectivity in a Pilot Generative Model of Zero-Degree Calorimeter Showers

Julian Juan

juliansjuan08@gmail.com

October 2026

Abstract

We test whether similar mean deposited energy and active-channel counts imply similar shower structure for a hierarchical neutron generator in an ePIC Zero-Degree Calorimeter simulation. The target is raw energy in 6,790 stored channels: every positive deposit counts as active, without digitization or an analysis threshold. One model checkpoint trained on 26,624 showers is compared with Geant4 at 10,000 matched incident conditions spanning 50–250 GeV kinetic energy. Among nonempty showers, the mean active-layer count differs by 0.25, yet the last layer with a deposit lies 4.42 layers farther downstream and the occupied span contains 3.95 more skipped layers. The mean number of connected hit groups on the model graph rises from 23.42 to 59.39. After allowing for empty layers that split the hit pattern into contiguous active-layer segments, at least 30.45 of the 35.97 excess groups (84.6%) remain within those segments. High-level classifier AUROC is 0.775, above the recorded 0.65 screen; a condition-only control gives 0.464 with row splits that can separate matched pairs. These aggregate results use one training seed and a repeatedly inspected validation sample. Threshold dependence, detector segmentation, repeated fits and reconstruction effects remain unmeasured.

1 Introduction

At the Electron–Ion Collider, a zero-degree calorimeter (ZDC) measures neutral particles near the outgoing hadron beam. A proposed ePIC SiPM-on-tile design reconstructs neutron energy and angle from hit energies and positions [1]. A surrogate’s spatial pattern could therefore affect reconstruction, which this study does not test. Here the hit pattern (mathematical support) comprises stored channels with positive deposit. Its size is the active-channel count, not digitized occupancy.

Calorimeter generators use flows, diffusion and flow matching on grids, point clouds and graphs [2–6]. CaloChallenge evaluates energy, shape, classifier and cost observables [7, 8]. ALICE ZDC studies include variational autoencoders, adversarial models, diffusion, flows and expert mixtures [9–13]; recent work tests output variability [14] and neighboring-channel co-activation [15]. Different detector representations and targets preclude direct numerical comparison with this raw-channel study.

We study a generator that samples total deposit, active layers and deposited-energy hit patterns in sequence. We ask whether similar mean deposits and hit counts imply similar depth and graph structure. They do not here: the last deposit occurs later, more layers are skipped, and many extra graph-connected groups remain after accounting for empty-layer separations. Their reconstruction effect is unmeasured; the diagnostic complements established co-activation tests.

This evaluation study uses one checkpoint and 10,000 validation conditions; it tests neither architecture superiority nor detector performance.

2 Simulation target and study population

2.1 Readout target and conditioning

The event condition is the incident-neutron four-vector $p^\mu = (E, p_x, p_y, p_z)$, expressed in GeV. The five network inputs are

$$c = (\log(1 + K_{\text{inc}}/100 \text{ GeV}), u_x, u_y, u_z, \log(E/\text{GeV})), \quad \mathbf{u} = \frac{\mathbf{p}}{\max(\|\mathbf{p}\|, 10^{-12} \text{ GeV})}, \quad (1)$$

Here $\mathbf{p} = (p_x, p_y, p_z)$ is the three-momentum, $K_{\text{inc}} = \max(E - m_n, 0)$, and $m_n = 0.93956542052 \text{ GeV}$. For $E \geq m_n$, the two energy features carry the same physical information because $E = K_{\text{inc}} + m_n$; both reflect this checkpoint's input convention. Direction features vanish at zero momentum; the clamp is numerical.

The target $\mathbf{Y} \in \mathbb{R}_{\geq 0}^{6790}$ is the raw, nondigitized energy deposited in each valid readout channel, with repeated hits to one ID summed in preprocessing. Geant4 supplies the transport [16]. The event total sums stored deposits; it excludes energy lost outside the readout. Every positive deposit counts as a hit: no MIP-scale threshold or additional time cut is applied, and the production integration window is unknown. Tiny deposits may change gaps and graph connectivity. Zero total deposit defines an empty shower; its physical cause is unknown.

2.2 Geometry and graph representation

The stored geometry has 400 electromagnetic-calorimeter (ECAL) IDs in layer 0 and 6,390 hadronic-calorimeter (HCAL) IDs in layers 1–64: 100 per layer through 63 and 90 in layer 64. ECAL centroids have 30.3 mm horizontal pitch; HCAL mean layer coordinates have median z spacing 24.9 mm and first-to-last span 1.57 m. The ECAL lies at $z = 35.73 \text{ m}$ in the stored frame (Fig. 1).

A channel is a stored ID. For 2,400 HCAL IDs, two to four positions share that ID; their unweighted centroid defines its graph node. This multiplicity does not establish electronic ganging without a production volume/electronics map. The cited design has individually instrumented staggered tiles [1, 17], but equivalence is unverified. Production materials and coordinate origin are undocumented, so layer index cannot be converted to radiation lengths X_0 or nuclear interaction lengths λ_I .

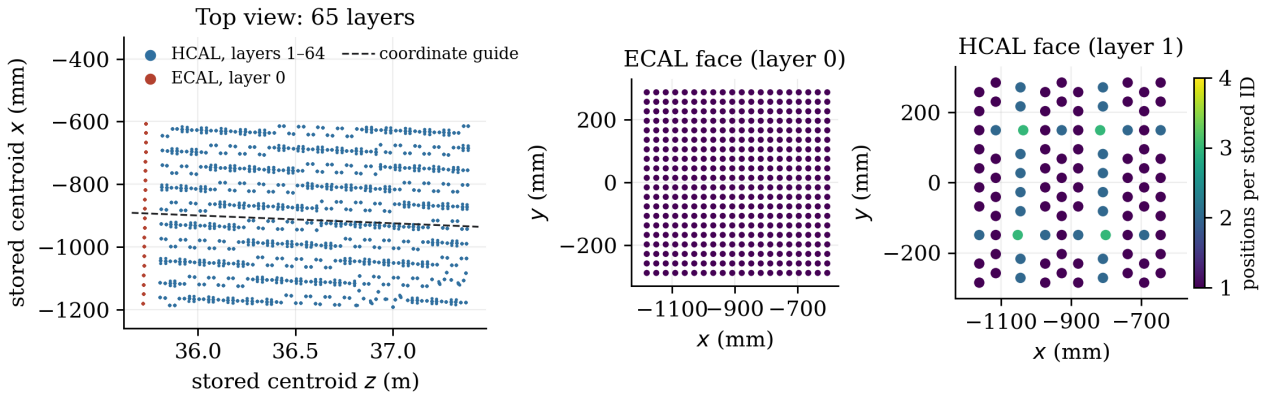


Figure 1: Stored-ID centroids. Left: horizontal projection of 65 layers, with a guide of $x/z = -0.025$, a 25-mrad slope comparable to the nominal EIC crossing angle (unequal axis scales). Center/right: ECAL/HCAL faces, colored by positions per ID. The stored coordinate frame does not establish beam-axis alignment; ID multiplicity does not establish electronic ganging.

The fixed graph gives each channel eight nearest-centroid links within its layer. For each channel in layers 0–63, it also selects four nearest centroids in the next layer, with each selected longitudinal pair stored in both directions. The resulting directed edge set contains $8 \times 6790 = 54,320$ lateral and $2 \times 4 \times (400 + 63 \times 100) = 53,600$ longitudinal edges, or 107,920 in total. The same three-dimensional nearest-centroid graph supplies model messages and connects positive-deposit channels for diagnostics; its links are not verified physical tile neighbors.

2.3 Training and evaluation populations

The preparation record contains 764,940 showers over 0–300 GeV. A deterministic event-hash split assigns 612,482 to canonical training, 76,158 to validation, and 76,300 to the nominal test partition. The selected checkpoint instead used a pilot subset of 26,624 training and 6,656 validation events; the pilot training population is 4.35% of canonical training.

The primary shower comparison uses 10,000 incident conditions drawn from canonical validation with $50 \leq K_{\text{inc}} \leq 250$ GeV. We refer to this repeatedly inspected validation subset as the development bank D_{dev} . For each four-vector, we compare one stored Geant4 shower with one independently sampled model shower. The pair shares its incident condition; its stochastic shower histories are independent. The development bank’s overlap with the 6,656-event pilot-validation subset cannot be reconstructed from the released aggregates, although the recorded split cross-tabulation places pilot training and validation events within their corresponding canonical partitions. No nominal test event is used here. This reanalysis uses the evaluation summary and static geometry. The publicly released summary lacks per-event masks and energies; exact contiguous-segment counts, paired structural intervals and threshold studies require another event-level evaluation with the collaboration-owned events and checkpoint.

3 Generator and training procedure

The generator builds a stored shower readout in physical order: total deposit, layer activity and energy, then channel positions and deposits (Fig. 2). It separates how much energy is stored (T), where along the detector it appears ($F, \mathbf{A}, \mathbf{B}$), and how it is distributed transversely ($\mathbf{K}, \mathbf{S}, \mathbf{Y}$). Counts precede positions, as in point-cloud models [5, 18]. These observables describe a shower outcome; the cascade does not simulate individual interactions or particle trajectories.

The model has 2,082,507 trainable parameters. A two-block residual multilayer perceptron encodes the five inputs in Eq. (1) into a 128-component condition vector h_c . Discrete heads generate visibility, layer activity, channel counts, and support; two conditional flow-matching models generate the continuous layer and channel energy shares from independent Gaussian draws. There is no learned shower encoder or common random event latent. At inference, the incident four-vector is the only event input. The embedding h_c encodes energy and direction for a fixed geometry; its learned coordinates have no assigned physical observables. Random draws represent conditional shower variability, not added electronics noise.

Below, $\ell = 0, \dots, 64$ indexes layers, i indexes valid channels in layer ℓ , and $Y_{\ell i}$ is a stored deposit. A channel is active if $Y_{\ell i} > 0$ and a layer if its budget $B_\ell = \sum_i Y_{\ell i} > 0$. The event total is $T = \sum_\ell B_\ell$.

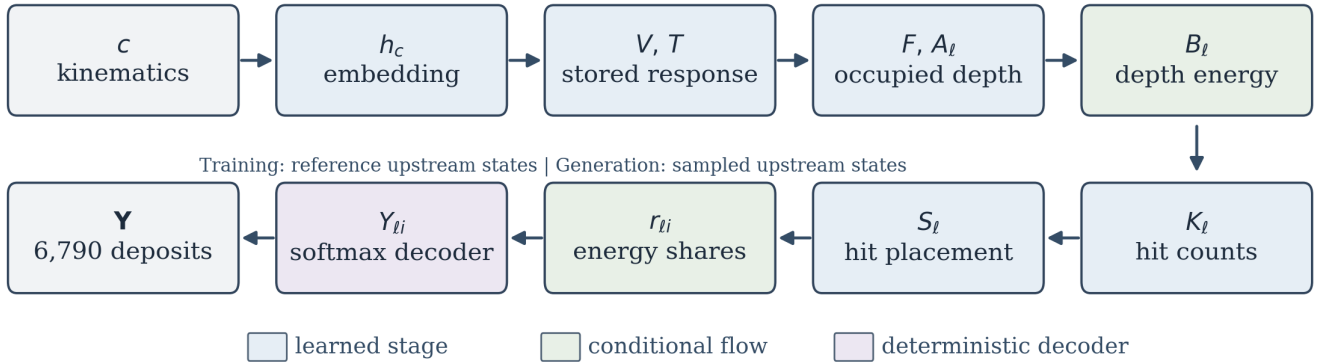


Figure 2: Read the top row left to right, then the bottom row right to left. The cascade maps incident kinematics to stored deposits. Green stages generate energy shares by conditional flows; purple decodes deposits. Arrows show generation order, not every conditioning connection or particle transport. The middle annotation contrasts training and generation inputs.

3.1 Total deposit and longitudinal allocation

Stored response. The two-part response stage first samples whether total stored energy is zero, then samples the magnitude when positive. We call the positive-deposit indicator visibility; it describes the stored readout, not whether a neutron underwent an inelastic interaction. A Bernoulli head samples a provisional visibility flag V_0 , while a four-component Gaussian mixture describes the transformed total q_T . The Bernoulli logit a_θ , mixture means μ_m , positive widths σ_m , and softmax weights ω_m depend on h_c ; θ denotes the learned parameters and σ the logistic function:

$$\begin{aligned} \Pr(V_0 = 1 \mid h_c) &= \sigma(a_\theta(h_c)), & q_T &= \log(1 + T/10 \text{ GeV}), \\ p_\theta(q \mid h_c) &= \sum_{m=1}^4 \omega_m(h_c) \mathcal{N}(q; \mu_m(h_c), \sigma_m^2(h_c)). \end{aligned} \quad (2)$$

A draw q is mapped back to deposited-energy units, clipped at zero, and capped at the training-derived value $C(K_{\text{inc}})$. The final visibility flag also requires positive incident kinetic energy and a positive candidate deposit:

$$\begin{aligned} C(K_{\text{inc}}) &= \min\{0.630110 \max(K_{\text{inc}}, 1 \text{ GeV}), 61.2338 \text{ GeV}\}, \\ T_{\text{cand}} &= \min\{10 \text{ GeV} \max(e^q - 1, 0), C(K_{\text{inc}})\}, \\ V &= V_0 \mathbf{1}_{K_{\text{inc}} > 0} \mathbf{1}_{T_{\text{cand}} > 0}, & T &= VT_{\text{cand}}. \end{aligned} \quad (3)$$

The cap saturates at 61.2338 GeV above $K_{\text{inc}} \approx 97.2 \text{ GeV}$. Its training-derived tail bound is a numerical safeguard, not a sampling fraction or a mean-response prediction. Clamping a nonpositive inverse-transform value to zero can produce an empty shower even when $V_0 = 1$.

Layer activity. For a visible event, the model draws the first active layer $F \sim \text{Cat}[\text{softmax } f_\theta(h_c, \log(1 + T/\text{GeV}))]$ from 65 categorical probabilities. It forces layer F active and all earlier layers inactive. Separate Bernoulli draws set the later indicators A_ℓ ; conditional on (h_c, T, F) , their joint probability factorizes as

$$p(\mathbf{A} \mid h_c, T, F) = \prod_{\ell > F}^{64} p(A_\ell \mid h_c, T, F, \ell), \quad A_F = 1. \quad (4)$$

F marks the first stored deposit, not necessarily the first inelastic interaction. The mask $\mathbf{A}$ specifies which layers contain any positive deposit. At fixed (h_c, T, F) , Eq. (4) samples later layers independently: an intermediate layer can be empty even when deposits occur on both sides. Physical cascades correlate deposits across depth, but whether this assumption causes the observed gaps requires a controlled test. The budget flow then assigns the longitudinal energy profile and ECAL/HCAL partition on that support. A star denotes a reference-derived quantity. For a reference shower, let $B_\ell^* = \sum_i Y_{\ell i}^*$, $T^* = \sum_\ell B_\ell^*$, and $A_\ell^* = \mathbf{1}_{B_\ell^* > 0}$. The target for a nonempty event is a centered log fraction on those layers:

$$a_\ell^* = \log \max\left(\frac{B_\ell^*}{\max(T^*, 10^{-12} \text{ GeV})}, 10^{-8}\right), \quad (x_1)_\ell = A_\ell^* \left(a_\ell^* - \frac{\sum_m A_m^* a_m^*}{\max(1, \sum_m A_m^*)}\right). \quad (5)$$

For an empty reference shower, the target and mask are zero. The centering removes a common log offset that the softmax ignores. Gaussian noise can contain this offset, which final normalization removes. Floors of 10^{-8} and 10^{-12} GeV keep the operations finite; they are numerical conventions, not detector thresholds, and alter small shares. Their activation frequency was not measured.

Energy allocation by flow matching. Both energy-allocation flows learn a velocity field that transports Gaussian noise toward the distribution of reference log fractions, conditioned on the incident particle and upstream shower state. Flow time is an artificial interpolation coordinate, independent of detector time. Straight-path conditional flow matching regresses noise-to-target displacements [19, 20]. For each training example, x_0 is standard-normal noise on the active coordinates, x_1 the reference-derived target, and t is uniform on $[0, 1]$. A mask $M_{bj} \in \{0, 1\}$ identifies active coordinate j of minibatch example b :

$$x_t = (1 - t)x_0 + tx_1, \quad \mathcal{L}_{\text{FM}} = \frac{\sum_{b,j} M_{bj} [v_\theta(x_t, t; \text{ctx})_{bj} - (x_1 - x_0)_{bj}]^2}{\max(1, \sum_{b,j} M_{bj})}. \quad (6)$$

The velocity field v_θ predicts the straight-path displacement $x_1 - x_0$ from the interpolated state x_t , time, and context. Inactive coordinates do not contribute to the masked squared-error loss.

During generation, both flows start from fresh Gaussian noise and use eight masked velocity updates. The context contains outputs sampled by earlier stages, whereas training supplies their reference values. For an active-layer or selected-channel mask M , the implemented rule is

$$z_0 = M \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad z_{n+1} = M \odot \left[z_n + \frac{1}{8} v_\theta \left(z_n, \frac{n+1/2}{8}; \text{ctx} \right) \right], \quad n = 0, \dots, 7. \quad (7)$$

Here $\odot$ denotes elementwise multiplication. Each update evaluates the velocity at the current state z_n and the midpoint *time* $(n + 1/2)/8$, then resets inactive coordinates to zero. It is a first-order explicit update with midpoint-time evaluation, not the midpoint Runge–Kutta method, which also advances the state to the midpoint. For a visible event, the resulting active-layer logits are normalized to fractions π_ℓ , giving budgets

$$\pi_\ell = \frac{A_\ell e^{z_{8,\ell}}}{\sum_{m:A_m=1} e^{z_{8,m}}}, \quad B_\ell = T\pi_\ell, \quad \sum_{\ell=0}^{64} B_\ell = T. \quad (8)$$

Only active layers enter the softmax denominator; their budgets sum to the sampled total and inactive layers receive zero. The layer-flow network uses a 128-component representation and two bidirectional transformer layers with four attention heads. Its context includes h_c , the total T , layer identity, the activity mask, and a sinusoidal flow-time encoding. Bidirectional attention lets each layer use information from all other layers; it is a statistical dependency, not backward particle transport.

3.2 Channel counts, hit pattern, and deposited energy

Channel counts. Given (h_c, B_ℓ, A_ℓ) and a learned layer identifier, a categorical head draws the number K_ℓ of active channels. A feasibility mask sets $K_\ell = 0$ for inactive layers and restricts active layers to $1 \leq K_\ell \leq N_\ell$, where N_ℓ is the number of valid channels. This models readout multiplicity at a given layer energy. It does not specify particle multiplicity, footprint area, channel positions or a minimum deposit.

Channel placement. The support network uses channel geometry and the sampled layer state to score candidate channels. Its inputs combine standardized (x, y, z) centroids, normalized layer index, ECAL/HCAL indicators, h_c , B_ℓ , and the requested fraction K_ℓ/N_ℓ . Three graph blocks of width 96 propagate information along the fixed directed edge set $\mathcal{E}$. Each block forms messages from neighboring readouts and uses their sum to update the receiving channel. If $h_i^{(r)}$ is the representation of channel i entering block r , and e_{ji} contains five edge inputs (three coordinate differences, distance, and edge type), one block computes

$$m_i^{(r)} = \sum_{j:(j,i) \in \mathcal{E}} \phi_r(h_j^{(r)}, h_i^{(r)}, e_{ji}), \quad h_i^{(r+1)} = h_i^{(r)} + \psi_r(h_i^{(r)}, m_i^{(r)}). \quad (9)$$

Here ϕ_r computes each message and ψ_r computes the residual update. Layerwise averages of the resulting representations pass through a two-layer, four-head bidirectional transformer; its outputs are broadcast back to channels before their final scores $s_{\ell i}$ are formed. Messages let scores depend on nearby readouts; they are statistical comparisons, not particle or energy transport. The support S_ℓ determines transverse placement and therefore graph connectivity. Selection is a separate stochastic step. For valid-channel set $\mathcal{N}_\ell$, the sampler chooses

$$S_\ell = \{i \in \mathcal{N}_\ell : \text{rank}_{\mathcal{N}_\ell}(s_{\ell i}/\tau + g_{\ell i}) \leq K_\ell\}, \quad (10)$$

Rank one denotes the largest noisy score. The sampler uses temperature $\tau = 1$ and independent Gumbel perturbations $g_{\ell i}$ obtained from uniform draws clipped to $[10^{-7}, 1 - 10^{-7}]$ [21]. Gumbel-top- K samples K_ℓ distinct channels without replacement. Selecting the top K_ℓ enforces the requested count exactly, but imposes no adjacency or connectivity constraint: isolated positive-deposit channels remain possible despite graph-informed scores.

Channel energies. A second flow divides each layer budget among its selected channels, describing energy concentration and fluctuations on fixed support. It can change energy-weighted shape and centroids, but cannot relocate deposits or repair disconnected support in exact arithmetic. In training, $S_\ell^\star$ denotes the positive-deposit reference channels and $K_\ell^\star = |S_\ell^\star|$. For $K_\ell^\star > 0$, its target is the centered log fraction of each channel’s layer budget:

$$a_{\ell i}^\star = \log \max \left(\frac{Y_{\ell i}^\star}{\max(B_\ell^\star, 10^{-12} \text{ GeV})}, 10^{-8} \right), \quad (x_1)_{\ell i} = \mathbf{1}_{i \in S_\ell^\star} \left(a_{\ell i}^\star - \frac{1}{K_\ell^\star} \sum_{j \in S_\ell^\star} a_{\ell j}^\star \right). \quad (11)$$

Here x_1 refers to the channel-flow target rather than the layer-flow target. The centering preserves the floored relative shares. An inactive layer has zero target coordinates. This flow uses Eqs. (6) and (7) with the reference channel set during training and the sampled set during generation.

The channel-flow velocity network follows the support network’s graph and layer-context design, with the flow state and a sinusoidal time encoding supplied at each channel. Unselected channels are masked before and after every graph block. Eight updates yield logits $r_{\ell i}$ for selected channels; a deterministic softmax then assigns the energy

$$Y_{\ell i} = \begin{cases} B_\ell \frac{\exp r_{\ell i}}{\sum_{j \in S_\ell} \exp r_{\ell j}}, & i \in S_\ell, \\ 0, & i \notin S_\ell, \end{cases} \quad (12)$$

Consequently, in exact arithmetic, $|S_\ell| = K_\ell$, $Y_{\ell i} \geq 0$, $\sum_i Y_{\ell i} = B_\ell$, and $\sum_{\ell i} Y_{\ell i} = T$. These are accounting identities for the model’s sampled readout budget, not conservation of the incident neutron’s full energy. Energy in unrecorded material and leakage are not generated; positivity and closure alone do not establish a physical shower.

3.3 Training and checkpoint selection

During training, each stage receives reference-derived upstream values (teacher forcing); when generating a full shower, it receives the preceding stages’ sampled values. Errors can therefore propagate through the cascade. We evaluate those complete generated showers in Sec. 5. The joint training objective combines nine weighted losses:

$$\begin{aligned} \mathcal{L}_{\text{joint}} = & w_V \mathcal{L}_{\text{BCE}}(V) + w_T \mathcal{L}_{\text{NLL}}(T) + w_F \mathcal{L}_{\text{CE}}(F) + w_A \mathcal{L}_{\text{BCE}}(\mathbf{A}) + w_B \mathcal{L}_{\text{FM}}^B \\ & + w_K \mathcal{L}_{\text{CE}}(\mathbf{K}) + w_S \mathcal{L}_{\text{BCE}}(\mathbf{S}) + w_R \mathcal{L}_{\text{rank}}(\mathbf{S}) + w_Y \mathcal{L}_{\text{FM}}^Y. \end{aligned} \quad (13)$$

Binary cross entropy (BCE) trains visibility, later-layer activity, and channel-support decisions; categorical cross entropy (CE) trains first-layer and count distributions. The total-deposit negative log likelihood (NLL) uses the density in transformed q_T . Expressing that density in deposited-energy units introduces the Jacobian factor $1/(T + 10 \text{ GeV})$. The ranking term averages $\log[1 + \exp(-(s_i - s_j))]$ over sampled occupied–unoccupied channel pairs. Superscripts B and Y denote the masked layer-budget and channel-share flow losses in Eq. (6); the count loss downweights zeros in inactive layers.

The retained calibration proposal uses training-data gradient norms to set w_j . In that proposal, response, layer-profile, and count weights hit the lower clipping bound; the visibility weight hit the upper bound. The bounds apply before normalization of the weights to mean one. Appendix A records the calibration values and their provenance limits. A weight-sensitivity study would test whether these bounds affect shower structure.

Optimization proceeded through response, profile, count, support, share, and joint stages. The condition encoder was trained with the response head, frozen for the intermediate component stages, and unfrozen for joint training. Each stage inherited the preceding checkpoint; freezing the encoder kept the inputs to previously trained heads fixed during intermediate stages.

Training used AdamW with a peak learning rate of 3×10^{-4} , batches of six accumulated over four steps (effective batch size 24), gradient clipping at one, and seed 20260723. Epoch 90 minimizes the teacher-forced validation objective across 104 retained rows spanning epochs 11–114 and defines the checkpoint analyzed below. Selection used this objective; the free-running support diagnostics were evaluated subsequently. The checkpoint need not minimize those free-running discrepancies.

4 Evaluation protocol

We compare total and ECAL/HCAL deposits, active-layer patterns, and positive-deposit channel patterns with Geant4. Response includes empty showers and eight 25 GeV energy bins. Numerical checks test the generated deposits against the model’s sampled budgets.

Hit-pattern measurements count every strictly positive stored deposit, with no detector or analysis threshold. For a nonempty shower, let F and L be the first and last layers containing a deposit, and A_ℓ their activity indicator. The occupied span is $L - F + 1$; the number of skipped layers inside it is

$$G = L - F + 1 - \sum_{\ell=0}^{64} A_\ell. \quad (14)$$

The longitudinal gap count G counts skipped layers, not gap regions; $G > 0$ marks an interior gap. Graph-connected hit groups are weak components $C_1, \dots, C_m$ of positive-deposit channels, ignoring edge direction. For active set S , the largest-group fraction is $\max_j |C_j|/|S|$. These are zero-threshold graph groups, not reconstructed topological clusters. Hit-pattern means use 9,907 reference and 9,858 generated nonempty showers, which need not share incident conditions.

Graph edges join only the same or adjacent layers. Let R count contiguous active-layer segments, each a maximal block with deposits in consecutive layers. Empty layers block paths between segments, so $m \geq R$. Consecutive empty layers constitute one separation; G alone does not determine R or within-segment disconnection. For example, layers 1, 2, 5 have $G = 2, R = 2$, while 1, 3, 5 have the same span and G but $R = 3$. Define $Q = m - R$, the extra graph-connected groups within contiguous segments (Fig. 4). If each segment forms one group, $Q = 0$; $Q > 0$ permits lateral splitting or failed links across adjacent active layers. Writing $I = \mathbf{1}_{G>0}$ gives $1 + I \leq R \leq G + 1$: any interior gap requires at least two contiguous segments. Thus $\bar{m} - 1 - \bar{G} \leq \bar{Q} \leq \bar{m} - 1 - f$, where $f = \bar{I}$ is the nonempty gap fraction. Subtracting the separate sample bounds gives

$$\Delta \bar{m} - \bar{G}_{\text{gen}} + f_{\text{ref}} \leq \Delta \bar{Q} \leq \Delta \bar{m} + \bar{G}_{\text{ref}} - f_{\text{gen}}, \quad \Delta = \text{generator} - \text{reference}.$$

This aggregate bound is neither a confidence interval nor causal attribution. Event masks would give exact R, Q , but were not retained. It discounts empty-layer separations but does not isolate transverse splitting. These graph-connected groups are not detector-reconstruction clusters [22].

For all-event active-channel counts, W_1 is the area between empirical cumulative distributions, in channels [8]. The 95% intervals for W_1 , zero-deposit fraction and mean normalized response use 1,000 paired bootstrap resamples stratified by energy bin. Intervals for the energy-bin differences, mean longitudinal profile, gaps, and graph components are unavailable. All quoted intervals describe resampling variation within the development bank.

A classifier two-sample test (C2ST) distinguishes generated from reference showers; AUROC (area under the receiver operating characteristic curve) has chance benchmark 0.5 [8, 23]. The three-seed classifier battery splits rows. Members of a matched condition pair can enter opposite partitions despite sharing the same incident four-vector. This permits condition memorization to favor the opposite held-out label. Its condition-only AUROC of 0.464 is a failed control consistent with such bias, though the cause is unverified. A separate pair-grouped monitor for `dicos-f-02`, epoch 90, records condition-only/high-level AUROCs 0.500/0.893. Its different event sample precludes a direct comparison. Appendix B gives evaluator details.

5 Results

Table 1 compares all-event deposits with hit-pattern statistics on separately nonempty showers. Similar mean deposits and active-layer counts coexist with longer occupied spans, more gaps and more graph-connected groups.

Table 1: Aggregate comparison on the development bank. Deposit means and zero-deposit counts use all 10,000 matched conditions; support means use nonempty showers (9,907 Geant4 and 9,858 generated). Differences are generator minus reference, computed before rounding; pp denotes percentage points.

Quantity	Geant4 reference	Generator	Gen. – ref.
Zero-deposit events	93 (0.93%)	142 (1.42%)	+49 (+0.49 pp)
Mean total deposit (GeV)	4.324	4.369	+0.045
Mean ECAL deposit (GeV)	2.100	2.275	+0.175
Mean HCAL deposit (GeV)	2.224	2.094	−0.129
Mean first active layer	0.510	0.735	+0.225
Mean last active layer	56.20	60.62	+4.42
Mean active layers	54.58	54.83	+0.25
Mean occupied span	56.69	60.88	+4.20
Mean inactive layers in span	2.10	6.05	+3.95
Events with an interior gap	52.43%	93.53%	+41.10 pp
Mean active channels	1617.6	1588.8	−28.7
Mean weak graph components	23.42	59.39	+35.97
Mean largest-component fraction	94.90%	88.87%	−6.03 pp

5.1 Deposited energy and empty showers

The all-event mean total deposit is 4.369 GeV for generated showers and 4.324 GeV for Geant4, a difference of 0.045 GeV (1.0%). Opposing section shifts are concealed: ECAL is higher by 0.175 GeV (8.3%) and HCAL lower by 0.129 GeV (5.8%) (Table 1). For an illustrative linear calibration $E_{\text{reco}} = c_E E_{\text{ECAL}} + c_H E_{\text{HCAL}}$, the shift is $\Delta E_{\text{reco}} \simeq (0.175c_E - 0.129c_H)$ GeV; unweighted cancellation need not persist. Coefficients and reconstructed energies are unavailable. Figure 3 shows the mean longitudinal allocation. At the reference HCAL maximum (layer 9), the generated mean is 8.6% lower. Mean profiles cannot reveal individual interior gaps.

Table 3 gives binwise means and widths; the pooled standard deviations are 4.759 and 4.830 GeV. The mean paired residual $(T_{\text{gen}} - T_{\text{ref}})/K_{\text{inc}}$ is 2.94×10^{-5} , with 95% bootstrap interval $[-8.51, 9.34] \times 10^{-4}$, which includes zero. This tests a different, energy-normalized quantity from the 0.045 GeV mean difference; it does not supply that difference’s paired covariance.

The model produces 142 empty showers versus 93 in Geant4, a difference of 49 events or 0.49 percentage points. The paired-bootstrap 95% percentile interval for this fraction difference is 0.19–0.76 percentage points. The corresponding nonempty samples contain 9,858 generated and 9,907 reference showers. The report’s visibility labels do not retain provisional V_0 , so the 142 generated zeros cannot be split reliably into hurdle and clamp contributions.

5.2 Longitudinal activity

The mean active-layer counts are close (54.83 generated versus 54.58 reference), but the last positive deposit is farther downstream. This extends the zero-threshold footprint; physical shower termination is unmeasured. Their mean first active layer shifts by +0.225 and last active layer by +4.42. The nonempty fraction starting in ECAL is 92.53% versus 93.81%. The occupied span grows by +4.20 layers while the count of active layers grows by only +0.25, leaving +3.95 inactive layers inside the span. The fraction of nonempty showers with at least one skipped layer rises from 52.43% to 93.53% (Table 1).

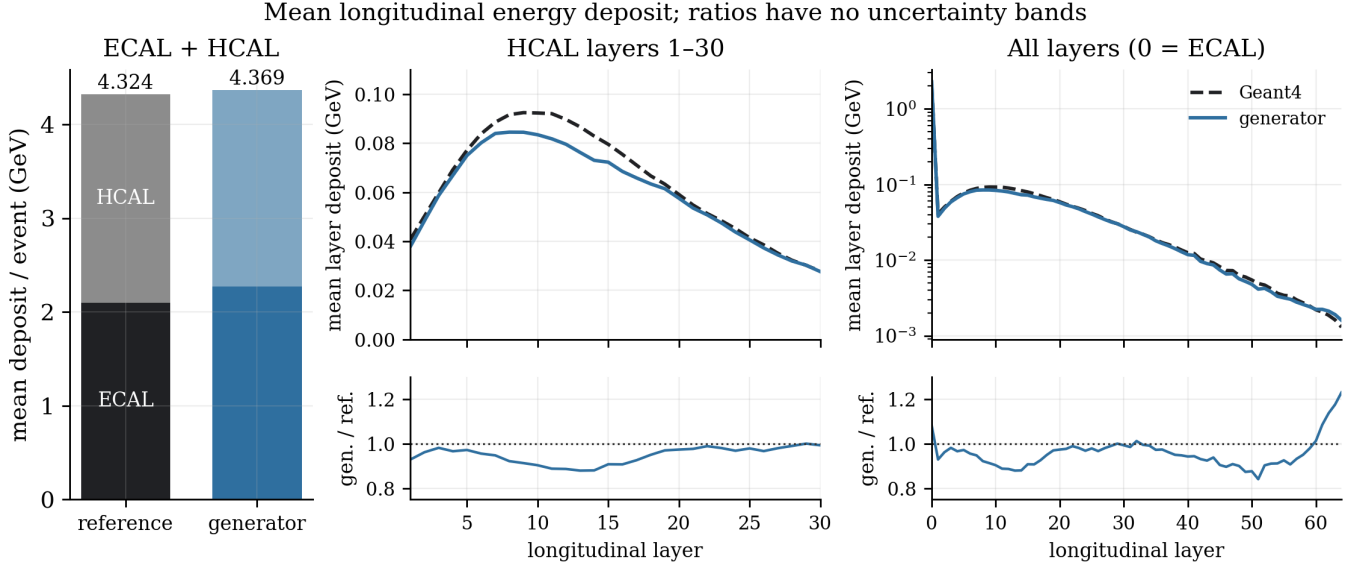


Figure 3: Mean deposited energy over all 10,000 development-bank events. Left: ECAL and HCAL deposits expose section-level compensation. Center: HCAL layers 1–30 (linear scale). Right: all layers, including ECAL at 0 (log scale). Dashed curves denote Geant4, solid curves the generator. Lower panels show ratios of means; uncertainty bands are unavailable. Empty showers are included.

For nonempty showers, mean energy-weighted depth, $\sum_{\ell} \ell B_{\ell} / \max(T, 10^{-9} \text{ GeV})$, is instead 14.14 for the generator and 14.15 for Geant4. Including empty events as zeros gives 13.94 and 14.02, respectively. This distinguishes occupied extent from energy-weighted depth, but does not determine the energy carried by disconnected components. Mean energy in layers 57–64 is 17.85 MeV for the generator and 17.19 MeV for Geant4, about 0.4% of each total. This fixed-region average does not isolate the energy of the additional occupied layers or disconnected components.

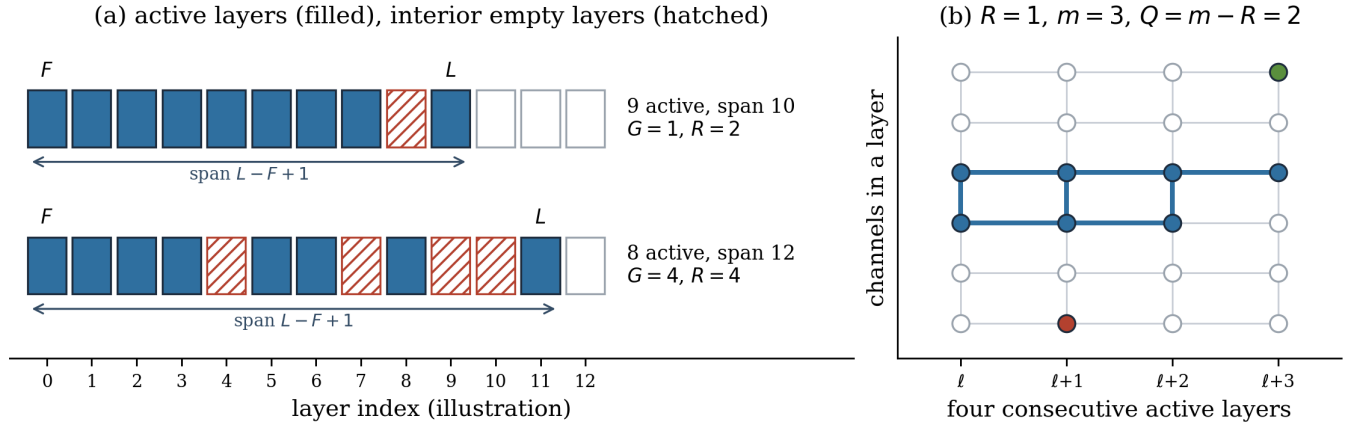


Figure 4: Definitions on an illustrative graph, not shower data. Left: active layers (filled), interior empty layers (hatched), first and last layers F, L , gap count G and contiguous active-layer segments R . Right: all four layers are active ($R = 1$), but occupied channels form three graph-connected groups ($m = 3$), so $Q = m - R = 2$. Pale edges join adjacent nodes of this toy grid; the detector graph is defined in Sec. 2.

5.3 Channel hit pattern and graph structure

Generated showers have 28.7 fewer positive-deposit channels but 35.97 more graph-connected hit groups; the largest group has 88.87% of occupied channels on average, versus 94.90% in the reference.

The bound leaves 30.45–37.14 excess groups within contiguous active-layer segments (at least 84.6% of the total).

The all-event active-channel-count distribution has $W_1 = 58.68$ channels, with a paired-bootstrap 95% interval of 51.47–66.92 channels. This distance includes the zero-count mass, whereas the means above condition on nonempty showers. Two retained reference-half splits give 13.78 and 15.82 channels (5,000 events per half). These are finite-sample comparison scales, not critical values or matched-size significance estimates.

The fraction of directed graph edges whose endpoints are both occupied averages 0.1375 for the generator and 0.1458 for Geant4 over all 10,000 events. This measures local co-occupancy on the same graph, but is not normalized for occupancy and does not isolate a correlation discrepancy.

5.4 Classifier and numerical checks

The original C2ST battery gives mean AUROCs of 0.775 for high-level shower summaries, 0.791 for channel energies and 0.856 for layer profiles across three classifier seeds. Its high-level score exceeds the recorded development-screening criterion of 0.65, whose pre-analysis status is not independently established. The failed condition-only control prevents a calibrated fidelity interpretation; a pair-grouped rerun on the same bank is needed. These scores and the failed criterion remain as provenance, not percentages of physically incorrect showers.

The evaluator uses a common event/layer residual tolerance: the larger of 2×10^{-5} GeV and 10^{-5} times the largest sampled event total in each batch. All 1,250 batches of eight passed, with no nonfinite/negative deposit or invalid support/count/mask. The zero support-mask mismatch verifies that FP32 decoding created no selected-but-empty channels in this sample; such losses would bias support metrics. Maximum event and layer residuals were 1.14×10^{-5} and 3.05×10^{-5} GeV. The maximum layer residual exceeds the earlier fixed 2×10^{-5} GeV criterion; the reported pass applies specifically to the batch-dependent criterion.

6 Discussion

6.1 Interpretation and diagnostic tests

The bound in Sec. 4 gives $\bar{Q}_{\text{gen}} \geq 52.34$ and $\bar{Q}_{\text{ref}} \leq 21.89$, hence $\Delta\bar{Q} \geq 30.45$: at least 84.6% of the 35.97-component excess remains beyond one group per contiguous active-layer segment. This establishes excess graph disconnection within contiguous active-layer segments. It need not be purely lateral: adjacent occupied layers may fail to connect. The bound assigns no causal share to a model stage; the separately nonempty samples need not share conditions.

Equation (4) assumes conditional independence of later-layer activity given (h_c, T, F) . In a three-layer example, fix $F = 0$ and let layers 1 and 2 be jointly active with probability p . Independent draws with the same marginals preserve the mean active-layer count but give an interior-gap probability $p(1 - p)$ instead of zero and increase the mean last layer by $p(1 - p)$. This explains a possible signature, not the measured magnitude. Sampled T and F can still induce dependence when their values vary; the first deposited layer is not necessarily the first inelastic-interaction depth.

A reference activity null controlling energy, direction, T and F would test whether factorization reproduces the depth and gap shifts. An oracle cascade must replace the full activity mask; reference T and F alone leave the factorization intact. L2LFlows and iCaloFlow model interlayer dependence [24, 25] and motivate a matched comparison.

BCE-trained logits are not automatically calibrated inclusion probabilities after fixed-count Gumbel selection. Independent channel perturbations can disrupt connected support even with smooth scores. Occupancy calibration conditional on layer and count, and a matched-marginal placement null, would test that hypothesis. The channel flow cannot repair selected support in exact arithmetic. Shared random upstream states and cross-layer attention can nevertheless induce centroid correlations without a common event latent; measuring those correlations would test shower-axis coherence.

6.2 Limitations and robustness

Paired sampling intervals for the headline structural means are not retained. Large descriptive differences do not establish robustness across training seeds, thresholds or graphs. A three-seed study and full-data fit are needed to separate architecture effects from training-population and optimization effects.

The strict $E > 0$ hit pattern describes stored simulation deposits, not electronics readout. Tiny or late deposits may change skipped-layer and connected-group counts; their contribution is unknown without channel-energy and timing data. Thresholding might remove a low-energy downstream tail or create new interior gaps; neither effect is measured. Threshold and timing scans are therefore necessary, together with an independently defined physical-neighbor graph, before interpreting this as a reconstruction defect. The cited design uses a 0.5 MeV MIP scale, a reconstructed-hit cut above 0.5 MIP and $t < 275$ ns [1]; these are design-study choices, not verified settings of the analyzed production. These motivate a 0.25 MeV HCAL sensitivity scan, but transferring the cuts requires the production segmentation and timing convention; a reference-only time cut would compare different targets.

The incident four-vector omits an explicit impact position. Whether direction determines that position depends on the production vertex and recording surface, which need verification. Position-conditioned residuals and the joint empty-shower contingency table would test whether conditioning and geometric acceptance are learned; marginal empty fractions alone cannot answer this.

Training loss averages updating minibatches, whereas validation uses the epoch-end model in the available implementation; historical runtime identity is unverified. Response-tail studies require counts of generated draws clipped by the cap and reference deposits above it. Numerical sensitivity requires a solver-step study. A locked bank with verified identities and repeated generation at fixed conditions is also needed.

6.3 Computational performance and detector applications

The evaluation took 3,442 s for 10,000 pairs. Its nonoverlapping timed analysis stages total 1,009 s, leaving 2,433 s (0.243 s per event) for generation, I/O and other untimed work; generation latency was not isolated. A matched end-to-end benchmark must include both solvers, decoding and output handling, with hardware and batch size specified for generator and Geant4.

Relating this raw-deposit study to detector performance requires production metadata, training-only calibration applied unchanged to both samples, and energy and position reconstruction. The ECAL and HCAL shifts cancel only in the unweighted stored-energy sum. A sampling fraction from another production cannot establish this model's calibrated bias.

7 Conclusion

This pilot comparison shows that similar mean deposited energy and active-channel count do not imply matched longitudinal or graph structure. A difference of 0.25 active layers coexists with a last active layer shifted by 4.42 layers and 3.95 additional inactive layers in the occupied span. At least 30.45 of the 35.97 extra graph-connected groups remain after accounting for contiguous active-layer segments. Empty-layer separations are therefore insufficient to explain the difference on this graph. Connecting this descriptive result to detector performance requires conditional activity and support-sampling tests, threshold and graph sensitivity, and reconstruction studies.

Code and data availability

The [project repository](#) provides aggregate reports, geometry, training history, figure scripts and manuscript source; its README identifies revisions and build audits. The collaboration-owned events and checkpoint are not redistributed. Generation, retraining and event-level studies require those resources.

Acknowledgments

The author thanks Dr. Wen-Chen Chang (Institute of Physics, Academia Sinica) for his generous mentorship, guidance, and encouragement, and the Institute of Physics laboratory community for their welcome and support.

Author contributions and use of generative AI

Julian Juan: conceptualization, methodology, software, investigation, visualization, and writing.

OpenAI Codex assisted with software and manuscript drafting, specification review, literature searches, and editing. Anthropic Claude provided an additional manuscript review and revision proposal; selected text and figure designs were incorporated after verification. The author supplied the research ideas, checked the work, and takes responsibility for the scientific content.

A Recorded settings and reproducibility limits

The analyzed checkpoint is `dicos-f-02`, epoch 90, from the `calibrated_lr3e4` family, with training seed 20260723. Its SHA-256 is

`491284c7423f365230d34b0443f95aa4888ec770bdc673c4c979897bad8acbce`

and its recorded frozen-configuration SHA-256 is

`116bc8c220b07ce54ae07196bdd6ed8e835775c8c937182a209a799dc94ae9c5`

The selected validation objective is 4.4837676194. A declared learning-rate restart separates epochs 70 and 71; the retained history covers epochs 11–114. The original runtime configuration, complete stage durations, other optimizer settings and exact training environment are unavailable in the paper package and cannot be inferred from aggregate losses.

Table 2: Recorded gradient-calibration proposal, rounded to six decimals and ordered as in Eq. (13). These values document the retained calibration record; the historical runtime configuration is unavailable for an independent check of the applied weights.

Weight	Component	Value
w_V	Visibility	2.574417
w_T	Total deposit	0.160901
w_F	First active layer	2.159451
w_A	Layer activity	0.536770
w_B	Layer-budget flow	0.160901
w_K	Channel count	0.160901
w_S	Support BCE	1.324108
w_R	Support ranking	1.477591
w_Y	Channel-share flow	0.444960

The retained weights use inverse median training-gradient norms on the shared encoder, clipped to $[0.25, 4]$ relative to their geometric mean and renormalized to mean one. Thus final Table 2 values can lie outside the clip range. This heuristic does not establish optimal weights.

The diagnostic report records generation seed 20260723 (the same integer as the training seed, used for a separate sampling run), FP32 precision on a CUDA device, batches of eight, and eight updates for each flow. Its validation-manifest identity is

`1bc3a6b21f736fc71551580a93207128c043833c6848627fb16f07475931690a`

The appendix tables use the same retained aggregate report as the main text; no new shower generation or test evaluation was performed to produce them.

The canonical split uses event hashes, not independent simulation jobs; this study uses validation only. Earlier separate studies inspected 40,000 nominal-test events and a draw containing 200 test events, with unresolved overlap. Those studies did not feed model decisions, but the nominal test partition is not wholly untouched. The retained record counts 36,100–36,300 uninspected events; their identities require recovery before use.

B Energy-bin results and classifier details

Table 3 resolves the aggregate response comparison into the eight incident-energy bins. The relative mean difference is $100(\mu_{\text{gen}}/\mu_{\text{ref}} - 1)$ and the width difference is $100(\sigma_{\text{gen}}/\sigma_{\text{ref}} - 1)$, where μ and σ are the mean and standard deviation of total deposited energy. Both statistics include empty showers.

Table 3: Deposited-energy response by incident kinetic energy. n is the number of matched conditions; the same n applies to each sample. Means and population standard deviations σ are in GeV; differences are percentages. Bins are lower-inclusive and upper-exclusive, except that the final bin includes 250 GeV (implemented upper edge 250.0001 GeV). Intervals for these differences are unavailable.

K_{inc} (GeV)	n	μ_{ref}	μ_{gen}	Mean diff. (%)	Width diff. (%)	σ_{ref}	σ_{gen}
50–75	1,182	2.398	2.216	−7.59	−6.40	2.830	2.649
75–100	1,270	2.892	2.908	+0.57	+2.80	3.154	3.243
100–125	1,310	3.527	3.513	−0.42	−0.72	3.872	3.845
125–150	1,226	3.849	4.146	+7.74	+9.58	3.921	4.297
150–175	1,288	4.867	4.494	−7.65	−10.89	5.037	4.488
175–200	1,268	5.030	5.320	+5.76	+8.79	4.802	5.224
200–225	1,266	5.921	6.026	+1.78	−1.70	6.127	6.023
225–250	1,190	6.095	6.330	+3.85	+5.24	5.827	6.133

Table 4: All recorded classifier seeds for the original development screening battery. The three seed columns vary the classifier and its row partition; they are not independent generator-training seeds. All three high-level AUROCs exceed the recorded screening maximum of 0.65.

Feature family	20260804	20260805	20260806	Mean
Condition only	0.4739	0.4624	0.4545	0.4636
High-level	0.7789	0.7719	0.7735	0.7748
Channel energies	0.7873	0.7943	0.7914	0.7910
Layer profile	0.8490	0.8580	0.8623	0.8564

The archived evaluator uses histogram gradient boosting (at most 100 iterations, depth four, learning rate 0.08). A class-stratified random split assigns 14,000 of 20,000 rows to fitting and 6,000 to scoring; library defaults may reserve fitting rows for early stopping. The seed controls both split and classifier. Pairs are not grouped. These folds lie within canonical validation, not the canonical train/test split.

High-level features are total deposit, active-channel count, energy-weighted depth, transverse centroids, radial RMS, largest-channel fraction, ECAL fraction and energy fraction in layers 48–64. Other families use all 6,790 deposits, 65 layer sums or incident kinetic energy alone. Total-energy denominators have a 10^{-9} GeV floor. Remaining classifier options follow library defaults; the historical library version is unknown.

The settings come from archived source, without a verified match to the historical runtime. The retained report records the 0.65 criterion, but its timing relative to analysis is unverified. Under the 70/30 row split, 42% of pairs straddle folds in expectation. Each scored row has a 70% chance of its identically conditioned opposite-label partner entering fitting (about 4,200 of 6,000 scoring rows in expectation). This permits a downward condition bias but does not establish the cause of AUROC 0.464. Table 4 cannot replace same-bank pair-grouped evaluation.

References

- [1] R. Milton et al. “Design and Simulation of a SiPM-on-Tile ZDC for the Future EIC, and Its Performance with Graph Neural Networks”. In: *Nuclear Instruments and Methods in Physics Research Section A* 1079 (2025), p. 170613. doi: [10.1016/j.nima.2025.170613](#). arXiv: [2406.12877](#).
- [2] C. Krause and D. Shih. “Fast and Accurate Simulations of Calorimeter Showers with Normalizing Flows”. In: *Physical Review D* 107 (2023), p. 113003. doi: [10.1103/PhysRevD.107.113003](#). arXiv: [2106.05285](#).
- [3] O. Amram and K. Pedro. “Denoising Diffusion Models with Geometry Adaptation for High Fidelity Calorimeter Simulation”. In: *Physical Review D* 108 (2023), p. 072014. doi: [10.1103/PhysRevD.108.072014](#). arXiv: [2308.03876](#).
- [4] D. Kobylanskii et al. “Graph-Based Diffusion Model for Fast Shower Generation in Calorimeters with Irregular Geometry”. In: *Physical Review D* 110 (2024), p. 072003. doi: [10.1103/PhysRevD.110.072003](#). arXiv: [2402.11575](#).
- [5] E. Buhmann et al. “CaloClouds II: Ultra-Fast Geometry-Independent Highly-Granular Calorimeter Simulation”. In: *Journal of Instrumentation* 19.04 (2024), P04020. doi: [10.1088/1748-0221/19/04/P04020](#). arXiv: [2309.05704](#) [[physics.ins-det](#)].
- [6] L. Favaro et al. “CaloDREAM: Detector Response Emulation via Attentive Flow Matching”. In: *SciPost Physics* 18 (2025), p. 088. doi: [10.21468/SciPostPhys.18.3.088](#). arXiv: [2405.09629](#).
- [7] C. Krause et al. “CaloChallenge 2022: A Community Challenge for Fast Calorimeter Simulation”. In: *Reports on Progress in Physics* 88.11 (2025), p. 116201. doi: [10.1088/1361-6633/ae1304](#). arXiv: [2410.21611](#).
- [8] R. Kansal et al. “Evaluating Generative Models in High Energy Physics”. In: *Physical Review D* 107 (2023), p. 076017. doi: [10.1103/PhysRevD.107.076017](#). arXiv: [2211.10295](#) [[hep-ex](#)].
- [9] J. Dubiński et al. “Machine Learning Methods for Simulating Particle Response in the Zero Degree Calorimeter at the ALICE Experiment, CERN”. In: *AIP Conference Proceedings* 3061 (2024), p. 040001. doi: [10.1063/5.0203567](#). arXiv: [2306.13606](#) [[cs.CV](#)].
- [10] M. Kita et al. *Generative Diffusion Models for Fast Simulations of Particle Collisions at CERN*. 2024. arXiv: [2406.03233](#) [[physics.data-an](#)].
- [11] M. Wojnar. “Even Faster Simulations with Flow Matching: A Study of Zero Degree Calorimeter Responses”. In: *Computer Physics Communications* 319 (2026), p. 109936. doi: [10.1016/j.cpc.2025.109936](#). arXiv: [2507.18811](#).
- [12] P. Będkowski et al. “ExpertSim: Fast Particle Detector Simulation Using Mixture-of-Generative-Experts”. In: *ECAI 2025*. 2025. doi: [10.3233/FAIA251042](#). arXiv: [2508.20991](#) [[cs.CV](#)].
- [13] M. Wojnar, E. Majerz, and W. Dzwiniel. “Fast Simulation of the Zero Degree Calorimeter Responses with Generative Neural Networks”. In: *Computing and Software for Big Science* 9 (2025), p. 1. doi: [10.1007/s41781-025-00130-x](#).
- [14] E. Majerz, W. Dzwiniel, and J. Kitowski. *Inverse Autoregressive Flows for Zero Degree Calorimeter Fast Simulation*. 2025. arXiv: [2512.20346](#).
- [15] E. Majerz et al. *Fine-tuned Normalizing Flows for ALICE Zero Degree Calorimeter Fast Simulation*. 2026. arXiv: [2608.12795](#) [[physics.ins-det](#)].
- [16] S. Agostinelli et al. “Geant4—a Simulation Toolkit”. In: *Nuclear Instruments and Methods in Physics Research Section A* 506 (2003), pp. 250–303. doi: [10.1016/S0168-9002\(03\)01368-8](#).
- [17] S. J. Paul and M. Arratia. “Leveraging Staggered Tessellation for Enhanced Spatial Resolution in High-Granularity Calorimeters”. In: *Nuclear Instruments and Methods in Physics Research Section A* 1060 (2024), p. 169044. doi: [10.1016/j.nima.2023.169044](#). arXiv: [2308.06939](#).
- [18] S. Schnake, D. Krücker, and K. Borras. *CaloPointFlow II: Generating Calorimeter Showers as Point Clouds*. 2024. arXiv: [2403.15782](#).
- [19] Y. Lipman et al. “Flow Matching for Generative Modeling”. In: *International Conference on Learning Representations*. 2023. arXiv: [2210.02747](#).
- [20] X. Liu, C. Gong, and Q. Liu. *Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow*. 2022. arXiv: [2209.03003](#).
- [21] W. Kool, H. van Hoof, and M. Welling. “Stochastic Beams and Where to Find Them: The Gumbel-Top- k Trick for Sampling Sequences Without Replacement”. In: *Proceedings of the 36th International Conference on Machine Learning*. Vol. 97. Proceedings of Machine Learning Research. 2019, pp. 3499–3508.
- [22] ATLAS Collaboration. “Topological Cell Clustering in the ATLAS Calorimeters and Its Performance in LHC Run 1”. In: *European Physical Journal C* 77 (2017), p. 490. doi: [10.1140/epjc/s10052-017-5004-5](#). arXiv: [1603.02934](#) [[hep-ex](#)].
- [23] D. Lopez-Paz and M. Oquab. “Revisiting Classifier Two-Sample Tests”. In: *International Conference on Learning Representations*. 2017. arXiv: [1610.06545](#).
- [24] S. Diefenbacher et al. “L2LFlows: Generating High-Fidelity 3D Calorimeter Images”. In: *Journal of Instrumentation* 18 (2023), P10017. doi: [10.1088/1748-0221/18/10/P10017](#). arXiv: [2302.11594](#).
- [25] M. R. Buckley et al. “Inductive Simulation of Calorimeter Showers with Normalizing Flows”. In: *Physical Review D* 109 (2024), p. 033006. doi: [10.1103/PhysRevD.109.033006](#). arXiv: [2305.11934](#).